

The dynamic complementarity tax

A problem statement (GO-13)

Geometric Observation program

Problem statement v0.7 — August 2026

Abstract

GO-10 measured the complementarity tax two consumers pay on one static source; GO-11 solved the exact region behind it, through the $m = 2$ frontier system with its 2×2 matrix water levels; GO-12 showed the staleness of the eraser's context is an access-width phenomenon, reduced entirely to one conditional variance $q_{\mathcal{G}}$. GO-13 asks the merged question: *what do two consumers pay when the eraser's context ages?* Nothing here is a claim; the conjectures are gates for future registrations.

1 The problem

Take GO-11's $m = 2$ setup verbatim (whitened latent $Z \in \mathbb{R}^{2+r}$, reads y_A, y_B , context block V now a stationary AR(1) *process* with observations $S_s = V_s + U_s$, $\text{Var } U = \tau^2$), single-letter two-dimensional records $\hat{\mathbf{Y}} = AZ + N$ with per-consumer distortions D_A, D_B . The eraser holds a context σ -algebra $\mathcal{G}$ from the observation process (slice, causal prefix, full path — GO-12's access family). Coordinates: the joint conditional coordinate $L_{AB}(\mathcal{G})$ (weighted-program value of GO-11 Theorem 9 with the conditional covariance induced by $\mathcal{G}$), the single-consumer coordinates $L_A(\mathcal{G}), L_B(\mathcal{G})$, and the taxes

$$\text{CT}_R = R_{AB} - \max(R_A, R_B), \quad \text{CT}_W(\mathcal{G}) = L_{AB}(\mathcal{G}) - \max(L_A(\mathcal{G}), L_B(\mathcal{G})).$$

CT_W is now a *function of the eraser's access*: the dynamic complementarity tax.

2 Conjectures

Theorem 1 (The matrix- q reduction; settled and netted, GO-P-2026-067). Hypotheses (all essential, per the R-IND-5 pass): joint Gaussianity; record noise $N \perp (T, \mathcal{G})$; the conditional-independence embedding $\text{Cov}(T, V_s) = \text{Cov}(T, V_t) a^{|t-s|}$; $\mathcal{G}$ generated by a nonempty subset of the observations $\{S_s\}$; weight $w \in [0, 1]$. Then for $m = 2$ single-letter records, $\mathcal{G}$ enters all coordinates only through $\Sigma_{T|\mathcal{G}} = \Sigma_T - (1 - q_{\mathcal{G}}) cc^T$ ($c = \text{Cov}(T, V_t)$, $q_{\mathcal{G}} = \text{Var}(V_t | \mathcal{G})$): the coordinates are GO-11 Theorem 9's program at $(\Sigma_T, \Sigma_{T|\mathcal{G}})$ — GO-12's channel-agnostic sufficiency argument transfers verbatim to 2-D records. Consequently, for *scalar* context every coordinate and both taxes are functions of $q_{\mathcal{G}}$ alone (**access-class universality**: equal- q classes give identical taxes — and only classes at *attainable* common q compare, each class having its own $q_{\min}$); the coordinates are nondecreasing in q (Loewner: $\partial_q \Sigma_{T|\mathcal{G}} = cc^T \succeq 0$, pointwise in every channel, feasible set q -free); and at $q \rightarrow 1$

they converge to the marginal program (continuity of the min from concavity of the value in q), so $\text{CT}_W \rightarrow \text{CT}_R$. Scalar universality *fails* for vector context $r \geq 2$ (verifier counterexample: equal $\text{tr Cov}(V \mid \mathcal{G})$ with coordinates differing by 0.12); the reduction survives only to the matrix $\text{Cov}(V \mid \mathcal{G})$. [R-IND-5: PASS, reduction to 3.6×10^{-15} , universality triples to 7×10^{-12} ; governed harness 067 ALL PASS 4/4, seed 20260921.]

Theorem 2 (The tax-curve characterization; settled and netted, GO-P-2026-068). Convention: q increasing means staler access. All of J, L_A, L_B are individually nondecreasing in q unconditionally (the q -sensitivity $f_q = (Ac)^\top M_1^{-1}(Ac) \geq 0$ pointwise), so only their difference can decrease. At any q where the optimum is attained, the solution set is locally bounded, and the context-coupling sensitivity is constant on the solution set (uniqueness of the optimal channel suffices, but is not necessary — a slack-constraint instance with a non-singleton argmin and constant sensitivity is on record), Danskin’s theorem [5] gives

$$\frac{dJ}{dq} = \frac{1-w}{2 \ln 2} (Ac)^\top M_1^{-1}(Ac), \quad \frac{dL_i}{dq} = \frac{1-w}{2 \ln 2} \frac{(u^\top c)^2}{Q_1 + n},$$

at the optimal channels (otherwise only the one-sided derivatives, $J'(q\pm) = \min/\max$ of the sensitivity over the argmin). Away from the max-switch locus, with the binding consumer any argmax of (L_A, L_B) (their sensitivities agreeing at ties),

$$\frac{d\text{CT}_W}{dq} = \frac{1-w}{2 \ln 2} \left[(Ac)^\top M_1^{-1}(Ac) - s_{\text{binding}} \right] :$$

staleness raises the tax iff the joint record’s context coupling, in the M_1^{-1} metric, exceeds the binding consumer’s. At an interior max-switch crossing the curve is continuous with a kink that is *always downward*: the right derivative subtracts the *larger* sensitivity (that consumer binds beyond q^*), the left the smaller, magnitude $\frac{1-w}{2 \ln 2} |s_A - s_B|$ (zero allowed at sensitivity ties; interior crossings require asymmetric distortions or variances). At $w = 1$ the curve is identically flat, and $\text{CT}_W \geq 0$ everywhere. [R-IND-5: PASS — 21 derivative checks to 1.5×10^{-9} , both signs realized, the kink’s one-sided slopes to 2.6×10^{-9} , the 067 instance’s reference-dependence confirmed (+0.177 max-based vs -0.0007 A -referenced); governed harness 068.]

Remark 1. Theorem 2 closes the question the 067 pass opened and explains its instances: I2’s increasing tax is a joint-coupling-dominant regime; the near-invariant curves arise when the binding consumer carries almost all the context coupling ($s_{\text{binding}} \approx s_J$). Open residue: the global regime map over $(\rho_{AB}, \rho_{AV}, \rho_{BV}, D_A, D_B, w)$ — for which the sign law is now the instrument — and the operational face (Question 3).

Remark 2 (The regime map — exploratory, unregistered, disclosed). An 18-point envelope sweep (`go13_regime_sweep.py`, seed 20260925; [exploratory]-grade only): the dynamic tax *rises* with staleness at 16/18 grid points; decreases are shallow slivers ($|d\text{CT}/dq| < 10^{-3}$) occurring only when the binding consumer is the context-*rich* one; when the binding consumer is context-poor the rise is strong at 10/10 points, up to +0.286 bits/unit- q . The candidate iff-law (“rises iff binding is context-poor”) *fails* at 6/18 points and is recorded as refuted-as-iff; the surviving asymmetry is the map. The operational face (draft registration 069) fixes its instances from this sweep: the grid-maximum rising instance, an equal- q universality control, and a paired flat contrast.

3 Questions

Theorem 3 (The spectral $m = 2$ theorem; settled and netted, GO-P-2026-073). Circulant embedding (essential: the verifier’s Toeplitz control breaks the mode mapping at $O(0.1)$; λ_k are the periodic AR(1) eigenvalues, mean 1); CI embedding with temporally *white* residual coupling (colored residuals change the mapping); jointly Gaussian linear records with two rows per symbol (hypothesis); noise $Q = I$ WLOG by scale invariance; bits/symbol, per-symbol mean distortions; $w \in [0, 1]$. Then the DFT modes are independent triples ($S_{A,k} = \rho_{AV}^2 \lambda_k + (1 - \rho_{AV}^2)$, cross-read covariance $(\rho_{AB} - \rho_{AV} \rho_{BV}) + \rho_{AV} \rho_{BV} \lambda_k$, context column $\propto \lambda_k$, conditional by the rank-one S -mode correction), cross-mode records gain nothing, and

$$J_{w,n}(D_A, D_B) = \min_{\substack{\frac{1}{n} \sum_k D_{A,k} = D_A \\ \frac{1}{n} \sum_k D_{B,k} = D_B}} \frac{1}{n} \sum_k V_k^{(9)}(D_{A,k}, D_{B,k}),$$

$V_k^{(9)}$ the static weighted matrix-water-level value of GO-11 Theorem 9 at mode k — **the static kernel is not ours**: the rate-face endpoint is Xiao–Luo (Allerton 2005) [3], and the static tuple’s reduction to pairwise subproblems with an implicit two-water-level system is Stylianou–Charalambous–Charalambous (ISIT 2021/2024) [4]; what this theorem adds is the *spectral lift*, the conditional/weighted face, and the third-party conditioner. At the allocation optimum the interior-mode partial slopes are equal per consumer: **two common prices** (μ_A, μ_B) , one per distortion constraint, generically distinct (a measure-zero coincidence set is possible) — this is where a “common pair of Lagrange prices” lives, vindicating GO-12 Conjecture 1’s phrasing at $m = 2$ after its $m = 1$ refutation. *Scoping*: equal slopes is KKT *necessity* (interior modes, complementary slackness for slack modes — a slack D_B collapses to the $m = 1$ one-price structure exactly, verifier-checked to 1.4×10^{-14}); *sufficiency* of the two-price rule is now PROVEN via Theorem 5 (whose convexity ingredients are standard [1, Sec. 3.1.7 and 3.2.4],[2]; the weighted program is convex in moments, $V_k^{(9)}$ jointly convex by perturbation, KKT sufficient given a Slater point; the allocation is globally optimal, unique under full-rank active records). Anchors: $n=1$ = static Theorem 9; $w=1$ = the two-constraint rate face; $\tau^2 \rightarrow \infty$ = the rate face; Toeplitz transfer inherits Theorem 1-lineage lemma debts. [R-IND-5: PASS — mode mapping 2.7×10^{-15} for $n \leq 8$; full-vs-allocation 7.4×10^{-14} over five settings and 100+ starts; prices $(-2.80690, -2.88527)$ with spreads $\leq 3.4 \times 10^{-7}$; five sharpenings folded. Novelty flank swept (record in the GO-13 novelty file): the spectral lift with two prices and the third-party conditioner remain claimable; language gated per the sweep.]

Theorem 4 (The binary twin; settled and netted, GO-P-2026-072). DSBS(p) pair at epoch t ; context a stationary symmetric binary Markov chain (per-step flip f , lag- k crossover δ_k); observations $S_s = V_s \oplus \text{Bern}(q)$; records $P(\hat{Y} | Y, V_t)$; access class $\mathcal{G}$ a *finite* subset of observations. Then: (i) **slice collapse is exact** — $\{S_{t \pm \Delta}\}$ reduces to Theorem 10 at $q_{\text{eff}} = q * \delta_\Delta$ (chain reversibility + BSC composition, both time directions); (ii) the σ -symmetric family remains optimal for every finite $\mathcal{G}$ (the joint flip fixes noise realizations pointwise, so the involution consumes no condition on q and covers asymmetric classes; convexity of the shared-channel conditional MI + symmetrization verbatim), with closed form

$$L_{\mathcal{G}}(d_0, d_1) = \mathbb{E}_r[h_2(a * r)] - (1 - p)h_2(d_0) - p h_2(d_1),$$

r the *posterior reliability distribution* of V_t given $\mathcal{G}$ — the per-atom step holds because $J = \hat{Y} \oplus V_t$ is *Bern*(a) *independent of* $(V_t, \mathcal{G})$ on the family (not by atom pairing; pairing belongs to the

symmetrization lemma alone); (iii) the **generalized tilt equation** $\ell(d_0) - \ell(d_1) = 2 \mathbb{E}_r[(1 - 2r) \ell(a * r)]$ is sufficient — the family is affine in (d_0, d_1) and MI is convex in the channel, so slice convexity is a *proof*, not an assertion; interior solutions may have $d_j \in (0, 1)$ (at $q = 0$, $d_1 \rightarrow 1$ as $D \rightarrow p$). (iv) **Universality is a Gaussian privilege**: no single q_{eff} reproduces a general class's curve — at $p=0.25$, $f=0.15$, $q=0.1$, class $\{S_{t-1}, S_{t+1}\}$, the one-point-fit gaps are $+7.1 \times 10^{-5}$ ($D=0.05$) and -8.9×10^{-5} ($D=0.15$), real at 50-digit precision — yet the family is *numerically near-universal*: over $D \in [0.02, 0.40]$ the probed envelope is $\leq 2.1 \times 10^{-4}$ (one-point fit) and $\leq 1.7 \times 10^{-4}$ (min-sup fit) — a numerical observation, not a theorem. Anchors: $\Delta=0$ slice = Theorem 10; $r \equiv \frac{1}{2} =$ marginal $1 - h_2(D)$; $q = 0$ regime $D \leq p$ confirmed. [R-IND-5: PASS — joints to 1.4×10^{-17} , closed form as an identity to 2.8×10^{-16} , direct optima on-family to 9×10^{-15} , non-collapse at 50 digits; six sharpenings folded, including the two numeric corrections now in this statement.]

Question 1. *The operational face: 058-lineage decode-threshold instruments with two consumers and an aged context, gating (i) the tax at two access classes tuned to equal q (the universality control), and (ii) the monotone tax curve across staleness levels. Instrument bias budget per PROTOCOL §5.1, with the paired-bias device of the 064 cross-net.*

Theorem 5 (The m -record moment-convexity lemma; Theorem 9's uniqueness resolved; settled and netted, GO-P-2026-074). For any $P \succeq Q \succeq 0$ and any record dimension m , on $\Omega = \{(C, V) : V - C^\top P C \succ 0\}$ the face $G(C, V) = \log \det(V - C^\top Q C) - \log \det(V - C^\top P C)$ is jointly convex: with $R = P - Q$ and $Z = R^{1/2} C M_Q^{-1} C^\top R^{1/2}$, $G = -\log \det(I - Z) = \text{tr } f(Z)$ with $f(z) = -\log(1 - z)$; M_Q is jointly matrix-concave, $C M^{-1} C^\top$ jointly matrix-convex and monotone (the classical matrix-fractional Schur certificate [1, Sec. 3.1.7 and 3.2.4], [2] — the block lift of the $m=1$ $aa^\top/s$ argument), so Z is jointly matrix-convex and $\text{tr } f$ finishes with scalar convexity alone. **Attribution**: both the scalar ($aa^\top/s$) and the matrix instances are *standard convex analysis* — the matrix fractional function is jointly matrix-convex and Loewner-nonincreasing in its denominator [1, Sec. 3.1.7 and 3.2.4], and the Schur-complement/operator-concavity ingredient is Ando [2]; *the program's contribution here is the lift into the moment chart, not the convexity* (GO-P-2026-074, attribution amendment dated 2026-08-07; no sealed gate, verdict or number changes). Hessian decomposition: $d^2 G = \text{tr}[(I - Z)^{-1} \ddot{Z}] + \sum_{ij} |\dot{Z}_{ij}|^2 / ((1 - \lambda_i)(1 - \lambda_j))$ — flats of G are exactly level flats of Z . Corollaries: the weighted $m=2$ program is CONVEX in moment coordinates (a sum of two instances); $V^{(9)}$ is jointly convex in (D_A, D_B) ; the optimal invariants (Z_1, Z_2) are unconditionally unique; (C, V) is unique for $w \in (0, 1]$, $S_1 \succ 0$, optimal C of full column rank (failure modes: a shut-off record — exactly the water-filling price-out — or a kernel of S_1 meeting the column space); the $\text{GL}(m)$ record gauge does not break constrained uniqueness (gauge orbits are curved, convex minimizer sets flat). GO-11 Theorem 9's uniqueness flag is resolved in this form. [Prover-grade R-IND-5: PROVED, four elementary steps, no operator convexity; identities and the Hessian decomposition verified to 10^{-13} ; seven instance families including $m > d_T$, zero violations.]

4 Verification obligations

House rules unchanged: R-IND-5 before assertion; novelty sweeps before novelty language (flanks listed in Question 1 and in go13-PLANNING.md); PROTOCOL §5.1 power/pilot at every seal; theorems harness-netted and CI-enforced. The first registration (GO-P-2026-067, sealed and run at v0.2) settled Theorem 1 with the equal- q universality control, and its R-IND-5 pass refuted the v0.1 monotone clause. The second registration (GO-P-2026-068, sealed and run at v0.3) settled the

tax-curve characterization as Theorem 2's envelope sign law, including the 067 label clarification as an instrumented gate. Open: the global regime map and the operational face.

References

- [1] S. Boyd, L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. (Sec. 3.1.7, the matrix fractional function; Sec. 3.2.4, composition rules.)
- [2] T. Ando, Concavity of certain maps on positive definite matrices, *Linear Algebra Appl.* 26:203–241, 1979.
- [3] J.-J. Xiao, Z.-Q. Luo, Compression of correlated Gaussian sources under individual distortion criteria, *Proc. 43rd Allerton Conf. Commun., Control, Comput.*, pp. 438–447, 2005. (Theorem 6 verified through the verbatim restatements in Nayak et al. 2008 and [4]; primary PDF not openly retrievable — see the novelty record.)
- [4] E. Stylianou, C. D. Charalambous, T. Charalambous, Joint rate distortion function of a tuple of correlated multivariate Gaussian sources with individual fidelity criteria, *Proc. IEEE ISIT*, 2021 (arXiv:2102.07236); explicit formulas in arXiv:2508.16301, 2025.
- [5] J. M. Danskin, The theory of max-min, with applications, *SIAM J. Appl. Math.* 14(4):641–664, 1966.